

A Guarded Single-Model Agent for CAR-bench: Deterministic Policy Enforcement for Consistent Tool Use

Anhtuan Le

WMG, University of Warwick

a.le.1@warwick.ac.uk

Abstract

CAR-bench [Kirmayr et al., 2026] credits a task under Pass³ only when all three trials succeed, so recurrent policy slips are amplified. In a public Qwen3-32B baseline, 62% of the tasks that fail Pass³ nevertheless succeed in at least one of five trials. We therefore make checkable behaviour invariant while reserving model-based deliberation for unresolved requests. The submitted system surrounds one tool-calling model with (i) a behavioural constitution compiled from the published policy, (ii) a transcript-derived intent ledger, and (iii) a deterministic pre-flight gate that withholds non-compliant calls and returns policy-grounded diagnostics. A fourth layer ranks three model-proposed plans after simulating their own effects. On one local surrogate stack, the guards raise Base trial-pass from 35/90 to 55/90; a paired task bootstrap gives a +22.2 point difference with 95% interval [+8.9, +36.7]. Equal-split composite Pass³ rises from .133 to .233, and selective deliberation reaches .267. Follow-up ablations on a reused 23-task stress subset are inconclusive. Absolute scores remain stack-dependent because the agent, user simulator and judge are local surrogate models.

1 Problem and Contribution

CAR-bench [Kirmayr et al., 2026] evaluates multi-turn, tool-using LLM agents acting as in-car voice assistants. Each task supplies 58 tools, 19 written policies and an LLM-simulated user, and the three task families (Base, Hallucination, Disambiguation) carry equal weight. The benchmark extends the tool-agent-user setting of τ -bench [Yao et al., 2024] with a strict consistency metric, since Pass³ credits a task only when all three trials succeed. An agent that solves a task twice out of three earns nothing for it.

Iterative reasoning methods such as ReAct [Yao et al., 2023] and Reflexion [Shinn et al., 2023] can raise per-trial capability. Pass³ nevertheless amplifies recurrent errors: under a simple independent-trial model, a task with per-trial success probability p contributes with probability p^3 . Our complementary strategy moves checkable policy behaviours from

probabilistic generation into deterministic code, while retaining model search where a request genuinely admits competing interpretations.

Which failures cost Pass³. A 100-task by 5-trial Base run of Qwen3-32B [Yang et al., 2025] at temperature 0 scores first-trial Pass¹ 0.62 with Pass@5 0.79. Only 21 tasks fail every trial, while 44 alternate between success and failure. Extra or missing state mutations, skipped mandatory reads and policy violations dominate the observed failures. A separate single-trial Claude Sonnet 4.5 profile has no tool-execution failures, so read coverage and state mutation are the primary targets, although the gate retains basic schema checks. The largest recurring trace pattern is a duplicate or corrective setter call, which writes an interim value and then adjusts it. This is fatal because every intermediate state hash must be a subset of the ground truth even when the value is later corrected.

2 Method

The submitted agent is one tool-calling model surrounded by three deterministic layers, with a fourth deliberative layer engaged selectively. Figure 1 shows the resulting loop, in which the model never addresses the environment directly, because a deterministic wrapper assembles the context, reads the proposed tool call, substitutes a ranked plan step where the request is ambiguous, validates the result, and only then allows a call to reach the car. No layer encodes task answers or reward internals.

L1, the compiled behavioural constitution. A build-time operating manual is appended to the evaluator’s system prompt. It restates the 19 policies as nine numbered procedures, covering read-before-write, one-shot actuation, the confirmation gate, a capability check against the current tool list, a disambiguation ladder, intent completeness, turn mechanics, speech style and cascade completion. The ladder fixes the order in which an underspecified request is resolved: strict policy, the explicit words of the user, stored preferences, policy defaults, current car state, and a question to the user only as the final rung. The same text compiles into the maps L3 enforces, namely 20 actuators mapped to their mandatory reads, the 26 state-changing tools, the 5 waypoint-edit tools that may never be batched, and four cascade postconditions.

L2, the intent ledger. Before every model call we derive from the message history alone which information tools have

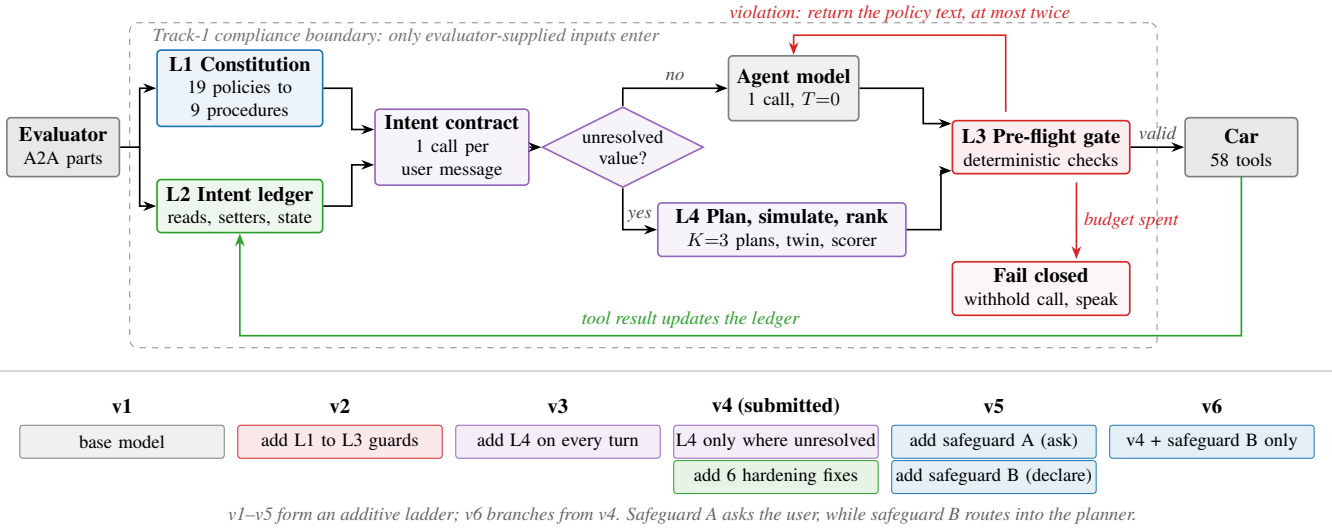


Figure 1: Architecture of the submitted agent, and the six versions evaluated in this report.

been read, which state-changing calls have succeeded, the canonical argument signature of each, the signatures that errored with nothing succeeding since, the last known vehicle state, and the cascade obligations outstanding. A success is committed on an observed non-error result and never on emission. The block enters the system prompt on every turn, so behaviour rests on a record rather than on the model re-reading a long transcript. Because the ledger is derived rather than written by the model, it is the one place in the loop where the account of what has happened is a fact, which is what allows the checks below to be deterministic.

L3, the pre-flight gate. Algorithm 1 gives the guarded loop. Well-formedness checks tool existence, valid JSON, required and unknown keys, primitive types, enumerations and nested objects; it is deliberately a limited schema subset rather than a complete JSON Schema implementation. Ordering requires a setter’s mandatory reads to be recorded in an earlier turn, an explicit affirmation for `REQUIRES_CONFIRMATION` tools, one waypoint edit per turn, and sequential policy-coupled changes. Idempotence blocks an identical state-changing call after success, or after an error with no intervening success. Value checks clear cascade obligations before speech, inspect returned state for the sunroof policy, consult preferences before asking or binding an unstated value, and reject ungrounded enum substitutions. The gate never rewrites a call. It returns a policy-grounded diagnostic for at most two model re-decisions, then withholds the call and emits one of eight safe speech templates. The enforcement logic is model-agnostic, although its end-to-end effect remains backend-dependent.

L4, selective plan, simulate and rank. One structured call parses the conversation into an intent contract whose entries carry tool-checkable postconditions. When a setter value is unbound or several readings remain, the model proposes $K=3$ plans under three strategy hints. An own-effect twin simulates them from ledger state, and a lexicographic scorer ranks policy violations, uncovered postconditions, parsimony and question count. Two deterministic speech candidates, of-

Algorithm 1 Submitted v4. M : history; T : tools; $B=2$ re-decisions.

```

1:  $\ell \leftarrow \text{LEDGER}(M)$ ;  $\ell.obl \leftarrow \text{CASCADE}(\ell)$   $\triangleright$  L2: derived
2:  $C \leftarrow \text{PARSECONTRACT}(M)$ ; cache its route for this user message
3: if  $C$  contains an unbound setter value or competing readings then
4:   generate  $K=3$  plans; simulate and rank them with two speech candidates
5:   queue the winner’s steps and hold its closing speech  $\triangleright$  L4
6: end if
7:  $S \leftarrow \text{wiki} \parallel \text{MANUAL} \parallel \text{RENDER}(\ell)$ ;  $G \leftarrow \emptyset$ 
8: for  $i = 0$  to  $B$  do
9:    $a \leftarrow$  next queued step, otherwise  $\text{LLM}(S \parallel M \parallel G, T)$ 
10:   $V \leftarrow \text{GATE}(a, T, \ell)$ ; if  $V = \emptyset$  then return  $a$ 
11:  if  $i = B$  then return  $\text{SAFETURN}(V, a)$   $\triangleright$  fail closed
12:   $G \leftarrow \langle a, \text{DIAGNOSTIC}(V) \rangle$ 
13: end for
14: on plan exhaustion: verify  $C$ ; otherwise replan once, then fall back

```

fering the remaining readings or declaring the request unavailable, compete alongside the model plans. The winner executes through L3, and closing speech is held until each postcondition is verified or declared impossible.

2.1 The six versions and the strategy behind each

Version **v1** is the bare model with the policy wiki and tool list. Version **v2** adds L1–L3, moving checkable rules from prose into code. Version **v3** adds L4 on every turn. Version **v4**, the submitted agent, restricts L4 to unresolved values and adds six trace-derived fixes: a success-only ledger; fail-closed exhaustion; value-aware cascades and relative writes; proof-carrying completion; route-before-parse with response-field refutation; and value fidelity, preference and idempotence checks. Version **v5** adds safeguard A, which asks once a value survives the preference rung, and safeguard B, which routes unsupported requests into the planner so that declaring unavailability competes with substitution. Version **v6** branches from v4 and adds B only; it was assessed only in the adaptive

stress follow-up.

Shipped configuration and cost. The submitted image is a self-contained uvicorn server implementing A2A [A2A Project, 2026], pinned for `linux/amd64`. Environment variables select model, provider, API base, key, temperature, thinking and reasoning effort. Completions pass through LiteLLM [BerriAI, 2026]; no provider is hard-coded. Defaults select v4, unresolved-value deliberation, $K=3$, response-field refutation, two gate re-decisions and both safeguards off. In the retained v4 records, the median final-turn prompt is 12.0k tokens and median LLM-induced latency per agent turn is 21.0s; local inference records zero API cost. The shipped instrumentation tracks finer call and token counts, but the archived benchmark results do not retain those counters, so we do not estimate total token cost from prompt snapshots.

3 Experimental Setup

No hosted-API credentials were available, so all three benchmark roles run locally on one NVIDIA L40S with 48 GB. The agent is qwen3:32b [Yang et al., 2025] in 4-bit quantisation, with thinking disabled, temperature 0 and a 32k context. The simulator and judge are qwen2.5-coder:14b [Yang et al., 2024], served by Ollama [Ollama, 2026]; the simulator runs at temperature 0.7 to provide repeated-trial variation. Absolute numbers are not leaderboard-comparable, so the analysis uses same-stack paired contrasts. Versions v1–v5 were developed adaptively and measured on the same 50-task training subset at 3 trials: 30 Base tasks stratified by baseline success plus 10 Hallucination and 10 Disambiguation tasks. The follow-up stress subset contains all 11 Hallucination tasks, 8 of 48 Disambiguation tasks and 4 of 17 Base tasks from a separately authored suite; v4, v5 and v6 use the same task and trial-index schedule. Because v6 was selected after observing v5 on this subset, that comparison is exploratory rather than a second independent held-out evaluation.

4 Results

Table 1 reports every version on one measurement stack, and two readings follow from it.

The first concerns the control layers. Moving from v1 to v2 raises Base trial-pass from 35/90 to 55/90. A paired bootstrap over the 30 tasks gives a +22.2 point difference with 95% interval [+8.9, +36.7]; a two-sided sign test has 12 tasks up and 3 down ($p=0.035$). Pass³ rises by .10 on each split and five of six reward-component failure counts fall; execution failures move from 16 to 17. This is the largest observed effect. The checks themselves are model-agnostic, but their end-to-end benefit is not. Restricting deliberation in v4 recovers most of what v3 gives away on Base while retaining its Disambiguation gain, producing the best training composite.

The second concerns the absolute level, which is a property of this stack: a 4-bit 32B agent and 14B simulator and judge. A manual taxonomy of 48 failures from an earlier guarded development run assigned 20 to model-capability errors, 8 to simulator artefacts and 20 to gaps in the control layers. This is diagnostic evidence, not a correction to Table 1: no counterfactual run establishes how many errors a different backend

Version	Pass ³ per split			Comp.	Trial	≥1
	Base	Dis.	Hal.	Pass ³	pass	of 3
v1 bare model	.300	.100	.000	.133	.360	26
v2 guards	.400	.200	.100	.233	.527	37
v3 L4 every turn	.333	.300	.000	.211	.487	36
v4 submitted	.300	.500	.000	.267	.513	38
v5 safeguards A+B	.233	.200	.000	.144	.467	38
Stress split	<i>n</i>		v4	v6	v5	
Hallucination	11		.303	.333	.242	
Disambiguation	8		.167	.250	.042	
Base	4		.667	.750	.917	
Pooled trials	23		.319	.377	.290	
Equal splits	3 splits		.379	.444	.400	

Table 1: Training version ladder (top) and adaptive stress follow-up (bottom).

Notes. Upper: 50 tasks at 3 trials per version; composite is the equal-weight split mean and the last column counts tasks solved at least once. Lower: trial-pass on the same 23 tasks and trial-index schedule. Pooled weights tasks; equal splits mirrors the benchmark’s split weighting.

would remove. The appendix fixes the scope of this audit and identifies the retained artifacts.

Version v4 solves 38 of the 50 tasks in at least one trial but converts only 14 into Pass³, leaving 24 mixed-outcome tasks under one configuration. Base Pass@3 of .800 against Pass³ of .300 measures the same consistency gap. Hallucination trial-pass ranges from 27% to 40% across v1–v5, showing that correct refusal is present but unstable. The v4 composite advantage over v2 comes from Disambiguation; because the ladder was tuned adaptively, its magnitude requires independent confirmation.

4.1 Follow-up ablations on a reused stress subset

On the training subset v5 scores below v4, at .144 against .267. On the reused stress subset, v5 lifts Base from .667 to .917 but lowers Disambiguation from .167 to .042. Its pooled rate falls from .319 to .290, whereas its equal-split mean rises from .379 to .400; neither summary supports a clear improvement. The failure records also do not justify attributing the entire drop to safeguard A: only 7 of 23 failed v5 trials carry DISAMBIGUATION_ERROR, so the observed split inversion remains a hypothesis-generating trace result.

Keeping safeguard B alone gives v6. Relative to v4, its pooled rate is +5.8 points with a task bootstrap 95% interval [−2.9, +15.9] and a sign test of six tasks up and three down ($p=0.51$). Its equal-split gain is +6.6 points with a split-stratified bootstrap interval [−5.2, +17.3]. Both intervals include zero, and v6 was chosen after inspecting v5 on the same subset. No complete retained v6 training run exists. We therefore keep v4 as the evidence-backed submission and leave both safeguards configurable but off by default.

Why the submission remains v4. The decision is based on evidence coverage, not on the largest point estimate. V4 has a complete 150-episode training record, the best equal-split training Pass³, and a direct comparison with every earlier architectural layer. V6 has the largest stress trial-pass estimate,

but only on the adaptively reused subset and without a retained training arm. At the stricter task level, v4, v6 and v5 achieve Pass³ on 3, 5 and 3 of the 23 stress tasks respectively. That ordering is compatible with a benefit from safeguard B, but five successes are too few to outweigh the missing replication and the intervals above. Retaining B behind an off-by-default switch preserves the hypothesis without making it part of the submitted claim.

Validity limits. Three dependencies bound the conclusions. First, the version ladder is exploratory: later mechanisms were designed from earlier traces, so its rows are not independent confirmatory experiments. Second, paired execution removes task-composition differences but not the local stack’s model, judge or simulator bias. The records preserve task and trial indices, not explicit random seeds, which prevents exact stochastic replay. Third, the stress families are deliberately unbalanced, making the pooled and equal-split summaries answer different questions. For that reason Table 1 reports both rather than choosing the favourable aggregation. The v1–v2 task-level interval is the strongest quantitative evidence; the version-specific Pass³ values and safeguard comparisons are descriptive until replicated on a frozen suite. The appendix documents the sensitivity calculation, run identifiers and image digest needed to audit that boundary.

4.2 Evidence Scope and Reproducibility

Interpretation. All results are same-stack contrasts, not estimates of official evaluation performance. The 30-task Base subset was stratified by baseline success, and versions v1–v5 were chosen iteratively on the 50-task training subset. The stress follow-up covers only 23 of 76 separately authored tasks: all 11 Hallucination, 8 Disambiguation and 4 Base. V6 was chosen after inspecting v5 on those tasks, so it supplies exploratory evidence only. “Pooled” in Table 1 weights each task equally; “equal splits” first averages within each task family and then gives the three families equal weight.

Sensitivity and failure audit. Between 9% and 17% of local trials contain a simulator-side anomaly such as an empty opening message or early stop. If such faults were independent of the agent and always fatal, a per-trial rate p would multiply Pass³ by $(1 - p)^3$, or 0.57–0.75 over this range. Those assumptions are not established, so we neither correct the scores nor claim a simulator-imposed floor. Separately, a manual taxonomy of 48 failures from the earlier guarded run `140s_guarded/base_train` labelled 20 model-capability, 8 simulator and 20 control-layer failures. The categories are diagnostic and do not predict performance with another backend.

5 Lessons Learned and Future Plan

What the traces taught us. Recording LLM calls, gate events and re-decisions turned scores into fixes. First, a deterministic layer is only as strong as the state it inspects. In one recurrent pattern, the gate correctly forced a precondition read, but checked only that the read occurred; the model then actuated despite the returned fan speed being zero. V4 therefore evaluates cascade predicates over ledger values, not tool names alone. Second, defaults decide outcomes. A blocked

high-beam call was proposed again after the re-decision budget even though the model’s discarded speech said to await confirmation; v4 withholds rather than defers to the invalid action. Third, prose is advisory where code is binding: promoting the preference rung into a deterministic check increased preference-tool use from 1/30 to 12/30 Disambiguation episodes in the relevant development comparison. These traces motivated the six v4 hardening fixes, but their benchmark effects are measured jointly rather than identified one by one.

What we would do differently. We over-trusted mechanisms that looked correct on development data. Failing closed after a prohibited action and safeguard A asking for an unresolved value can produce similar surface behaviour while obeying different policies; adversarial personas expose the distinction. We also optimised mean trial-pass more often than Pass³, while the 24 mixed-outcome v4 tasks needed more trials and trace-level labels. Finally, the stress subset should have been frozen before choosing v6, with random seeds and all internal counters retained in each result record.

Future plan. Three items follow from the ceiling in Table 1. The first is to treat Hallucination as a consistency problem rather than a knowledge problem, since the correct refusal already appears on roughly one trial in three; the step is to promote refutation into a pre-speech assertion gate, generalising the shipped response-field check from result fields to whole capability claims. The second is to extend the proof-carrying close to every split, so that no completion speech is emitted while a stated postcondition remains unverified. The third is to make the stack cheap enough to deliberate more widely. The constitution and wiki form a constant prefix of about 12k tokens re-sent on every step; placing that prefix before volatile ledger state enables provider-side prefix caching. The resulting savings should be measured as cumulative input and output tokens, latency and calls per completed episode before increasing K , rather than inferred from final-turn prompt size.

6 Conclusion

CAR-bench turns occasional policy errors into task failures, making deterministic control a useful complement to model reasoning. On our local stack, compiling the policy into a ledger and pre-flight gate produces the campaign’s clearest paired improvement; selective plan ranking then gives v4 the best equal-split training composite. The result is narrower than a leaderboard claim: absolute scores depend on three surrogate models, the ladder is adaptive, and the safeguard follow-up is too small and reused to resolve its six-point estimates. The submitted choice therefore follows completeness of evidence rather than the largest observed number. A frozen, seed-recorded replication across model backends is the next test of whether the guarded architecture transfers and whether safeguard B should become a default.

References

[A2A Project, 2026] A2A Project (2026). Agent2agent (A2A) protocol. <https://github.com/a2aproject/A2A>. Software, accessed 2026-07-28.

- [BerriAI, 2026] BerriAI (2026). LiteLLM: Call 100+ LLMs using the OpenAI input/output format. <https://github.com/BerriAI/litellm>. Software, accessed 2026-07-28.
- [Kirmayr et al., 2026] Kirmayr, J., Stappen, L., and Andre, E. (2026). CAR-bench: Evaluating the consistency and limit-awareness of LLM agents under real-world uncertainty. In Liakata, M., Moreira, V. P., Zhang, J., and Jurgens, D., editors, *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 40599–40618, San Diego, California, United States. Association for Computational Linguistics.
- [Ollama, 2026] Ollama (2026). Ollama. <https://github.com/ollama/ollama>. Software, accessed 2026-07-28.
- [Shinn et al., 2023] Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., and Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*.
- [Yang et al., 2025] Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., et al. (2025). Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.
- [Yang et al., 2024] Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., et al. (2024). Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.
- [Yao et al., 2024] Yao, S., Shinn, N., Razavi, P., and Narasimhan, K. (2024). τ -bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*.
- [Yao et al., 2023] Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. In *Proceedings of the International Conference on Learning Representations*.